H.T.No. | | | | | | | | | | |

Code No: CE4525 **SRGEC-R23**

III B.Tech I Semester Regular Examinations, November 2025

GREEN BUILDINGS

(Open Elective-I)

Time: 3 Hours **Max. Marks: 70**

Note: All questions from **PART-A** are to be answered at one place.
Answer one question from each unit of **PART-B.** All questions carry equal marks.

PART-A

10 × 2 = 20M

1. a) Summarize the essential sustainable features of a green building.
   b) Differentiate between conventional building materials and green building materials used in India.
   c) Explain how the LEED India Rating System promotes energy efficiency in buildings.
   d) Interpret the significance of water efficiency in green building practices.
   e) Outline the steps involved in reducing energy demand during building design.
   f) Illustrate the role of onsite sources and sinks in sustainable building performance.
   g) Explain the importance of energy modeling in HVAC system design.
   h) Compare the functions of chiller selection and cooling tower selection in green building design.
   i) Describe the measures taken to achieve acceptable indoor air quality levels.
   j) Assess the role of recycled-content materials in material conservation.

PART-B

5 × 10 = 50M

UNIT-I

2. a) Analyze the environmental benefits of green building adoption. (5M)
   b) Evaluate requisites for constructing sustainable buildings. (5M)

(OR)

3. a) Explain the benefits of green building materials. (5M)
   b) Explain the green building equipment in India. (5M)

UNIT-II

4. a) Assess the opportunities of green buildings in residential areas. (5M)
   b) Justify the impact of market transformation on green practices. (5M)

(OR)

5. a) Explain the role of IGBC in promoting sustainability. (5M)
   b) Illustrate features of material and resource management. (5M)

UNIT-III

6. a) Explain eco-friendly captive power generation for industries. (5M)
   b) Discuss the various steps to reduce energy demand. (5M)

(OR)

7. a) Demonstrate steps to maximize system efficiency. (5M)
   b) Interpret the role of renewable onsite sources. (5M)

UNIT-IV

8. a) Demonstrate how design interventions improve HVAC efficiency. (5M)
   b) Examine the role of interior lighting in green buildings. (5M)

(OR)

9. a) Evaluate precooling of fresh air in AC design. (5M)
   b) Explain the process of pump selection in HVAC systems. (5M)

UNIT-V

10. a) Interpret the causes of poor indoor air quality. (5M)
    b) Propose measures to control Sick Building Syndrome. (5M)

(OR)

11. a) Assess the significance of tobacco smoke control in IAQ. (5M)
    b) Evaluate the role of IAQ measurement in green performance. (5M)

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